India Impact Graph: Complete Project Brief

Hackwave 3.0 (Indore) | PS 5: Autonomous Financial Event Market Impact Graph | Version 1.0, 10 Oct 2026

How to use this document

This file explains the whole project: the problem, the solution, the data, the rules and the MVP. Give it to any teammate or any AI tool (ChatGPT, Claude, Gemini, Copilot, Cursor and so on) before asking for help, so that it understands what we are building and what we are NOT building.

Labels used: **[PS]** = written in the official problem statement. **[VERIFIED]** = confirmed from a public source. **[VERIFY]** = must be confirmed from the source before use. **[DECISION]** = our team's choice.

0. Hinglish summary (for the team)

News alag-alag dikhti hai, par asli me ek khabar ka asar chain me phailta hai. Jaise kaccha tel mehnga hua, to diesel, transport, sabzi, mehngai, byaaj dar aur loan sab pe asar padta hai. Hum ek system bana rahe hain jo news padhta hai, ek graph banata hai ki asar kahan tak phailega, aur end me batata hai ki India ke ek typical ghar (state, rural/urban, kharche ki class) ke mahine ke kharche pe kitna asar pad sakta hai, rupees ke range me. Har number ke saath official source ki receipt hoti hai. AI sirf news padhta aur explain karta hai. Numbers actual data aur formula se aate hain. Hum price predict nahi karte, hum samjhate hain ki asar kahan tak ja sakta hai aur kitna pakka hai.

1. The problem

1.1 Official problem statement [PS]

Financial events rarely affect a single company. A geopolitical event, commodity shock, interest-rate decision or regulatory change can propagate through sectors and financial instruments. Many systems show news as isolated stories rather than connected chains of impact.

The challenge: build an AI system that converts unstructured financial information into a dynamic dependency graph, showing how an event may influence economic variables, sectors and assets.

Illustrative chain from the PS: crude oil rises 12%, inflation expectations increase, interest-rate expectations shift, banks and real estate may be affected, relevant market indices and assets respond.

1.2 What the PS expects [PS]

- Detect financial events from unstructured information.
- Extract companies, commodities, sectors, indicators and instruments.
- Identify and score relationships between entities.
- Construct and update a knowledge graph.
- Estimate plausible propagation paths and affected assets.
- Incorporate time-series data where available.
- Explain each impact chain and the evidence behind it.

Technical challenge [PS]: combine NLP, event extraction, knowledge graphs, time-series data, graph algorithms and LLM-based reasoning. Distinguish observed facts from inferred relationships, and communicate uncertainty rather than presenting predictions as guarantees.

Demo expectation [PS]: introduce a financial news event and show the system extracting entities, building a connected graph, tracing multiple propagation paths, identifying potentially affected assets and explaining why each connection exists.

1.3 What the PS does NOT specify

Target user, market (India or global), tech stack, AI model, dataset, accuracy target, platform and judging criteria are not specified. Do not assume them. Our choices are listed in section 3.

1.4 The problem in simple words

A reader sees "crude oil up 12%", then "bank shares fall", then "real estate slows" as three unrelated stories. In reality they are linked. The reader has to work out the chain alone, and has no idea how it touches their own household. We want to make the chain, the evidence and the personal impact visible.

2. Our solution

2.1 One-line description

A news article goes in. The system extracts the event, builds a graph of how the effect spreads (solid line = observed, dashed line = inferred), shows 3 propagation paths with evidence, and shows the approximate monthly impact in rupees on a typical Indian household (a persona), with an official source receipt behind every number.

2.2 Pitch line

From global news to your household budget. Every number comes with an official receipt, and AI only reads and explains, it never does the maths.

2.3 What makes us different

Most teams will build: news, then an LLM, then a generic graph (oil, inflation, banks, stocks). Our differences:

1. **Receipt-backed numbers.** Every number on screen is clickable and shows its source, table and year. If data is missing, the screen says "Data not available". We never invent a number.
2. **Persona Impact Card in rupees.** Instead of abstract percentages, we show the approximate extra monthly spending for a typical household defined by state, rural/urban and expenditure class.
3. **News Consensus Meter.** For one event we compare 3 news sources: how many confirm it, where their claims conflict. This separates observed from inferred at the news level.
4. **AI-off proof.** A toggle turns the LLM off and the numbers stay the same. This proves AI is used only as reader and explainer.

3. Scope decisions [DECISION]

Item	In the MVP
Market	India
Levels	World, India, State, Persona

Item	In the MVP
City level and real individuals	Not included. No official data exists, and privacy would be a concern
Income	Not used. The official survey measures spending (MPCE), so we say "expenditure class"
Persona	3 states (Madhya Pradesh plus 2 more) x Rural/Urban x 3 expenditure classes (low, middle, high)
Items	8 to 10: petrol, diesel, LPG, edible oil, wheat/cereals, pulses, sugar, vegetables, electricity, gold
Events	2: crude oil +12% (main demo), plus a backup (edible oil or wheat spike)
News input	3 saved (cached) articles per event. No live news feed
Not in the MVP	City level, real person data, live news stream, login, more than 5 states, voice input, any number generated by AI

4. Data foundation

4.1 Verified facts

- **[VERIFIED]** The Ministry of Statistics (MoSPI) moved the Consumer Price Index (CPI) base year from 2012 to 2024. The new basket uses the Household Consumption Expenditure Survey (HCES) 2023-24, and the item count rose from 299 to 358.
- **[VERIFIED]** The new CPI series has data from January 2026. The old series (2012 base) runs up to December 2025. The two cannot be compared directly. MoSPI published linking factors and a back series.
- **[VERIFIED]** The HCES 2023-24 factsheet is available at state level and broad item-group level, and also by fractile (class) of monthly per capita consumption expenditure (MPCE).
- **[VERIFIED]** In HCES 2023-24, food is about 47.04% of rural spending and 39.68% of urban spending.

- **[VERIFY]** The new CPI food weight is reported as about 36.75% by one blog. Confirm from the official MoSPI table before using it.

4.2 Data sources to use

Need	Source	Status
Item weights	MoSPI CPI (2024 base)	[VERIFY] exact file
Spending by state, rural/urban, class	HCES 2023-24 (MoSPI)	[VERIFY] exact tables
Global commodity prices	World Bank Pink Sheet, FRED, FAO	[VERIFY]
India fuel prices	PPAC	[VERIFY]
India retail price history	MoSPI, RBI	[VERIFY]

4.3 Data rules

- Every data row stores its source and year.
- Do not mix the old CPI series (2012 base) with the new one. If the old series is used (for example in a backtest), label it "old series" clearly.
- If data for an item or group is missing, show "Data not available". Never fill gaps with estimates produced by AI.
- Keep "reach" (how many households use an item) separate from "impact" (how much budget is affected). Do not merge them into one number.

5. How the system works

5.1 Pipeline

1. **Input.** The user pastes or selects a news article.
2. **Reader (LLM).** Extracts the event type, magnitude (for example +12%) and entities, in a fixed JSON format.
3. **Base knowledge graph (curated).** Pre-built, sourced links such as crude oil to diesel, diesel to freight, inflation to RBI rate. These come from our data, not from the LLM.

4. **Path finder (graph algorithm).** Finds and ranks propagation paths from the event to affected nodes, using edge confidence.
5. **Household impact calculator (formula).** Computes the rupee impact range for the chosen persona.
6. **Explainer (LLM, controlled).** Writes the reason for each edge and path, using only the evidence and numbers given to it.
7. **UI.** Interactive graph, edge evidence panel, persona card, receipts, time-series chart, AI-off toggle.

5.2 Who does what (golden rule)

Task	Done by
Weights, spending patterns, prices	Actual data (CSV/JSON)
Impact calculation	Formula, not AI
Path finding and ranking	Graph algorithm
Reading news and writing explanations	LLM (reader and explainer only)

The LLM must never generate a number, never add a relationship and call it fact, and never explain with information that was not provided.

5.3 Graph design

- **Node types:** Commodity, Item (household), Indicator, Sector, Instrument/Asset, Persona.
- **Edge kinds:** Observed (written in the article, solid line) and Inferred (system's estimate from our base graph or LLM suggestion, dashed line).
- **Visual encoding:** edge thickness = impact strength, node size = budget share.
- **Edge click panel shows:** reason, evidence text, source, confidence, and "what could be wrong".

5.4 Main example path set

- Crude oil, then diesel, then freight, then food and vegetables, then household budget.
- Crude oil, then petrol, then commute cost, then household budget.

- Crude oil, then inflation expectations, then RBI rate expectations, then banks and real estate, then market indices.

5.5 Impact formula

`Persona Impact (Rs per month) = Sum over items ( item spending in Rs x price change % x pass-through )`

- **Item spending:** persona MPCE x the item's spending share (HCES/CPI data).
- **Price change:** from the event magnitude and pass-through.
- **Pass-through:** how much a global price move shows up in Indian prices, estimated from historical data for 3 to 4 items only [VERIFY]. For other items use "Data not available" or a sourced fixed value labelled "estimated".
- **Output:** always a range (low to high), never a single exact number, and never a guarantee.

5.6 Data schema (illustrative structure, not real data)

```
{
  "node": {"id": "crude_oil", "type": "commodity", "label": "Crude oil",
"source": "<source>", "year": "<year>"},
  "edge": {
    "from": "crude_oil", "to": "diesel",
    "kind": "observed | inferred",
    "direction": "positive | negative",
    "strength": "<number from data>",
    "confidence": "<low | medium | high>",
    "evidence_text": "<sentence from article or data note>",
    "source": "<source>", "year": "<year>"
  },
  "persona": {"state": "Madhya Pradesh", "area": "rural | urban", "class":
"low | middle | high", "mpce": "<from HCES>", "item_shares": {}}
}
```

6. MVP features

P0: required for the PS demo

1. News input and entity extraction (event, magnitude, entities in JSON).

2. Graph with solid (observed) and dashed (inferred) edges.
3. Three propagation paths with evidence on click.
4. Persona Impact Card in rupees, always as a range.
5. Receipt on every number: source, table, year, or "Data not available".

P1: our differentiators

6. Time-series chart (crude oil against India fuel prices or CPI, 1 to 2 series).
7. AI-off toggle (LLM off, numbers unchanged).
8. News Consensus Meter (3 sources compared).

P2 and P3

- P2: backtest on a past spike (using old series, clearly labelled), second event, state comparison, "what to watch" notes (not financial advice, rule-based and sourced).
- P3: export, dark mode.

7. Build order

1. Data files with source and year columns (weights, MPCE, 3 to 4 price series).
2. LLM extraction from news to JSON, tested on the 3 saved articles.
3. Graph and paths from static data for the crude oil event.
4. Persona Impact Card.
5. Receipts and observed/inferred styling.
6. Time-series chart and AI-off toggle.
7. News Consensus Meter.

Get steps 1 to 5 working end to end first. Do steps 6 and 7 only when 1 to 5 are stable.

8. Demo script (3 minutes)

1. State the problem: news is shown as separate stories, not as a chain.
2. Enter "Crude oil 12% up".
3. Entities are highlighted as observed.
4. Show the graph and 3 paths, open the evidence on one edge.

5. Show the persona card, for example a rural Madhya Pradesh middle-class household, as a rupee range.
6. Click a number to show its receipt.
7. Switch the AI-off toggle and show that numbers do not change.
8. Close: we do not predict, we explain how far an effect can travel and how sure we are.

9. Rules for any AI tool reading this file

1. Treat this document as the source of truth for project scope. Do not widen the scope.
2. Never invent data, statistics, weights or prices. If a value is unknown, mark it "Data not available" or "[VERIFY]".
3. Keep observed facts and inferred links clearly separate in code, data and UI.
4. Never present an impact as a guarantee. Always show a range and a confidence level.
5. Do not make the LLM do calculations. Calculations use the formula and the data files.
6. Do not add features outside section 6 unless the team asks.
7. Do not use real individuals' data. Personas are typical households.
8. Use "expenditure class", not "income".
9. Do not mix old (2012 base) and new (2024 base) CPI data.
10. Prefer a simple, working, demo-ready solution over a complex one.

10. Risks and fallbacks

Risk	Fallback
Pass-through data is noisy	Compute for 3 items only, others show "Data not available"
News API or scraping fails	Use the 3 saved articles per event and say they are cached
State data is not granular enough	Stay at broad item groups, do not show false precision
Old and new CPI series mismatch in backtest	Backtest only on the old series, labelled clearly

Risk	Fallback
LLM produces a wrong link	Common links come from the curated base graph, LLM suggestions are marked "Inferred, low confidence"
Scope creep	No P1 work until P0 is complete

11. Likely judge questions and short answers

- **Why AI?** Free-text news needs language understanding. AI only reads and explains, and the AI-off toggle proves the numbers do not depend on it.
- **How accurate is it?** We show ranges and confidence, not guarantees. Numbers come from official data, and we show a backtest on a past event (old series).
- **Where does the data come from?** Official sources (MoSPI CPI and HCES, commodity price series). Every number has a clickable receipt.
- **What if data is missing?** The screen says "Data not available". We never fill gaps.
- **How is this different from other teams?** Receipt-backed numbers, rupee impact on typical households, news consensus, and a proven non-blind use of AI.
- **Can it scale?** The pipeline is event-agnostic. More items, states and events only need more data rows.

12. Glossary

- **Event extraction:** pulling out what happened (for example "crude oil +12%") from text.
- **Entity:** a named thing such as a commodity, sector, company or indicator.
- **Knowledge graph:** nodes (things) connected by edges (relationships).
- **Propagation path:** the route an effect travels, for example oil to diesel to freight to food.
- **Observed vs inferred:** observed is written in the source, inferred is our estimate.
- **CPI:** Consumer Price Index, India's main retail inflation measure.
- **HCES:** Household Consumption Expenditure Survey, the official survey of household spending.
- **MPCE:** Monthly Per Capita Consumption Expenditure.
- **Pass-through:** how much of a global price change reaches Indian prices.

- **Persona:** a typical household defined by state, rural/urban and expenditure class.
- **Receipt:** the source, table and year behind a number, shown on click.

13. Open items

- Time left in the hackathon and who handles data, backend and frontend.
- The 2 extra states for personas (besides Madhya Pradesh).
- News sources for the 3 saved articles per event.
- Exact CPI 2024 weights file and HCES item-wise tables to download.
- Price series for pass-through (World Bank Pink Sheet, PPAC, MoSPI) and their availability.